

Shanika Iroshi Nanayakkara

Quantum Machine Learning | Quantum Federated Learning | Quantum Software and Optimisation

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LinkedIn | Google Scholar | GitHub | DeepUnfolding-Agent

Application: Postdoctoral Research Fellow - Spin System Quantum Computing Architectures

Position ID: SPIN001

RESEARCH PROFILE

PhD researcher in computer science and artificial intelligence with a doctoral thesis submitted for examination at Deakin University. Research expertise spans quantum machine learning, quantum federated learning, deep-unfolded optimisation, quantum-aware aggregation, communication-efficient learning, secure distributed AI, and research-software engineering. Experienced in developing reproducible Python/Qiskit software, conducting experiments with Qiskit Aer and IBM Quantum hardware, publishing in peer-reviewed journals and IEEE conferences, and teaching programming and software-engineering subjects. Seeking to apply transferable quantum-software, system-modelling, and adaptive-optimisation expertise to fault-tolerant silicon-spin architectures, compilation, and resource estimation.

CORE EXPERTISE

- Quantum computing: parameterised quantum circuits, variational quantum algorithms, quantum neural networks, quantum federated learning, IBM Quantum execution, quantum-noise analysis, shot-budget analysis, and circuit-depth evaluation.
- Quantum software: Qiskit, Qiskit Aer, Qiskit Machine Learning, SamplerQNN, EstimatorV2, SPSA and COBYLA optimisation, simulation, reproducible QPU workflows, and measurement-resource analysis.
- Machine learning and optimisation: federated learning, deep unfolding, FedProx, adaptive aggregation, non-IID learning, bilevel/meta-optimisation, validation-based model selection, and uncertainty-aware evaluation.
- Programming and scientific computing: Python, Java, C#, JavaScript, PHP, SQL, NumPy, SciPy, pandas, scikit-learn, PyTorch, Jupyter, and LaTeX.
- Software engineering: REST API development, Docker, Git/GitHub, reproducible experiment pipelines, testing, technical documentation, Linux, and cloud-based development.
- Academic communication: journal writing and revision, peer review, conference presentation, interdisciplinary collaboration, laboratory teaching, assessment, and technical mentoring.

EDUCATION

Doctor of Philosophy (Information Technology) - Deakin University, Australia **2021-present**

- Thesis submitted for examination in July 2026: "Improving and Modeling Federated Aggregation." Degree pending examination and conferral.
- Research areas: quantum and classical federated learning, quantum machine learning, deep-unfolded optimisation, adaptive aggregation, communication efficiency, and security.
- Deakin University Postgraduate Research Scholarship recipient.

Master of Information Technology (Software Design and Development) - Charles Sturt University, Australia **2019-2021**

- Postgraduate University Medal (2021), Executive Dean's Award (2020), Academic Achievement Award (2019), and Academic Merit Award (2019).

BSc (Hons) in Information Technology - University of Moratuwa, Sri Lanka **2013-2017**

- Second Class Upper Division; GPA 3.58/4.20.

ACADEMIC AND RESEARCH APPOINTMENTS

Graduate Research Teaching Fellow and Sessional Academic - Deakin University, Australia **2024-2026**

- Delivered and supported Information Technology tutorials and laboratories, including programming and Android application development, student consultation, assessment, marking, and progress-oriented technical support.
- Translated programming, software-engineering, machine-learning, and research concepts for students with varied technical backgrounds.

Research Assistant - Deakin University, Australia **2025**

- Contributed to research implementation, experimentation, technical documentation, reproducible code, and research-support activities.

Graduate Student Instructor - University of Moratuwa, Sri Lanka**2017-2019**

- Taught and supported programming, software engineering, databases, object-oriented analysis and design, web technologies, image processing, and operating systems.
- Provided laboratory instruction, assessment support, consultation, and student project guidance.

Visiting Lecturer - Multimedia Technologies and Graphic Design - OpenArc Campus, Sri Lanka**2018**

- Delivered lectures and practical sessions and supported student project work.

RESEARCH SOFTWARE AND SELECTED PROJECTS

DeepUnfolding-Agent - Lead Researcher and Developer. Developed a strategy-agnostic API for deep-unfolded adaptive control in classical, quantum, and hybrid optimisation. Implemented controller creation, state-conditioned control suggestions, controlled quantum rounds, autonomous experiments, persistence, Docker deployment, simulation support, and IBM Quantum integration. [Project website](#) | [GitHub repository](#)

Deep-Unfolded Quantum Federated Learning. Developed controller-driven QFL methods using adaptive SPSA learning-rate and perturbation controls, proximal regularisation, validation-based best-unfold selection, and outer meta-optimisation; evaluated using Qiskit Aer and IBM Quantum hardware.

Semantic-Aware Post-Quantum Security Module. Proposed a ML-KEM-based security mechanism that adapts cryptographic protection according to the significance and communication context of federated model updates. Integrated post-quantum key establishment with authenticated encryption to protect model exchanges against replay, stale-model, client-substitution, and message-manipulation attacks.

Quantum-Aware Aggregation and Communication. Designed fidelity-aware angular aggregation, A2G-QFL, stable aggregation, predictive-residual, aggregation-impact, and teleportation-assisted communication methods for heterogeneous and noisy quantum federated systems.

HantaWatch: Federated Genomic Surveillance. Developed a privacy-preserving federated framework for Hantavirus genomic surveillance, combining source-aware client construction, adaptive optimisation, model selection, uncertainty flags, and expert-review prioritisation.

Classical Deep-Unfolded Federated Optimisation. Developed adaptive FedProx-style controllers that learn local learning rate, proximal strength, and training effort from optimisation-state features under client and data heterogeneity.

RESEARCH PROJECT DELIVERY AND COLLABORATION

- Managed end-to-end thesis-by-publication workstreams spanning problem formulation, algorithm design, theoretical analysis, data preparation, simulation, real-QPU evaluation, software engineering, manuscript preparation, revision, and dissemination.
- Coordinated reproducible experiments across multiple datasets, heterogeneous client configurations, quantum simulators, and IBM quantum processors; maintained versioned code, dependency specifications, experimental logs, and technical documentation.
- Worked across quantum computing, machine learning, communications, cybersecurity, bioinformatics, and software-engineering contexts, and communicated research to academic, technical, and general audiences.

PUBLICATIONS**Peer-Reviewed Journal Articles**

1. **S. I. Nanayakkara**, S. R. Pokhrel, and G. Li, "Understanding global aggregation and optimization of federated learning," *Future Generation Computer Systems*, vol. 159, pp. 114-133, 2024. doi: [10.1016/j.future.2024.05.009](#).
2. **S. I. Nanayakkara** and S. R. Pokhrel, "Modeling Deep Unfolded Quantum Machine Learning Framework," *IEEE Transactions on Artificial Intelligence*, 2025. doi: [10.1109/TAI.2025.3573303](#).

Peer-Reviewed Conference Papers

3. **S. I. Nanayakkara** and S. R. Pokhrel, "Fidelity-Aware Angular Aggregation for Quantum Federated Learning over Noisy Entanglement," *Proceedings of the 40th International Conference on Information Networking (ICOIN)*, 2026. doi: [10.1109/ICOIN68469.2026.11480586](#).
4. **S. I. Nanayakkara** and S. R. Pokhrel, "A2G-QFL: Adaptive Aggregation with Two Gains in Quantum Federated Learning," *Proceedings of the 2026 International Conference on Quantum Communications, Networking, and Computing (QCNC)*, Kobe, Japan, pp. 808-812, 2026. doi: [10.1109/QCNC69040.2026.00132](#).
5. **S. I. Nanayakkara** and G. U. Ganegoda, "Biometric Traits in Enhancing Security of Healthcare," *Proceedings of the 3rd International Conference for Convergence in Technology (I2CT)*, 2018. doi: [10.1109/I2CT.2018.8529442](#).

Manuscripts Under Review or Revision

1. **S. I. Nanayakkara** and S. R. Pokhrel, "Quantum Federated Aggregation: Advances and Problems," *ACM Computing Surveys*, submitted 2026. Under review. Preprint: [10.5281/zenodo.21348244](#).

2. **S. I. Nanayakkara** and S. R. Pokhrel, "HantaWatch: Federated Learning for Hantavirus Genomic Surveillance," *IEEE Journal of Biomedical and Health Informatics*, submitted 2026. Under review. Preprint: [10.5281/zenodo.21351899](https://zenodo.org/record/21351899).
3. **S. I. Nanayakkara** and S. R. Pokhrel, "A Stable Aggregation Method for Quantum Federated Learning," *IEEE Journal on Selected Areas in Communications, Quantum Communications and Networking Series*, submitted 2026. Under review. Preprint: [10.5281/zenodo.21351842](https://zenodo.org/record/21351842).
4. **S. I. Nanayakkara** and S. R. Pokhrel, "A Drift-Stable Quantum Federated Learning for Intelligent Services," *IEEE Transactions on Services Computing*, submitted 2026. Revision 1 under review. Preprint: [10.5281/zenodo.21351623](https://zenodo.org/record/21351623).
5. **S. I. Nanayakkara** and S. R. Pokhrel, "Communication-Efficient Deep Unfolded Quantum Federated Learning," *IEEE Transactions on Networking*, submitted 2026. Under review. Preprint: [10.5281/zenodo.21351787](https://zenodo.org/record/21351787).
6. **S. I. Nanayakkara** and S. R. Pokhrel, "Modeling Secure and Efficient Quantum Federated Learning of Vehicular Networks," *IEEE Transactions on Mobile Computing*, submitted 2026. Under review. Preprint: [10.5281/zenodo.21352024](https://zenodo.org/record/21352024).
7. **S. I. Nanayakkara** and S. R. Pokhrel, "DeepUnfolding-Agent: A Strategy-Agnostic API for Deep-Unfolded Adaptive Control in Classical, Quantum, and Hybrid Optimization," *Future Generation Computer Systems*, submitted 2026. Under review.

SELECTED PRESENTATIONS AND CONFERENCE LEADERSHIP

- ICOIN 2026, 40th International Conference on Information Networking, Vietnam - presented thesis-related research on fidelity-aware aggregation for quantum federated learning.
- IEEE QCNC 2026, International Conference on Quantum Communications, Networking, and Computing, Kobe, Japan - presented research on adaptive aggregation in quantum federated learning.
- ATIS 2025, International Conference on Applications and Techniques in Information Security - presented thesis-related work on federated learning, quantum-aware modelling, and intelligent systems.
- Session Chair, QUICK 2025, Vellore Institute of Technology, India.
- Deakin University School of Information Technology Three-Minute Thesis Competition, March 2023 - presented the doctoral research and its broader significance to a general academic audience.

TEACHING AND SUPERVISION-RELATED EXPERIENCE

- Laboratory teaching and assessment in Android application development, programming, software engineering, databases, web technologies, image processing, and operating systems.
- Technical mentoring of undergraduate and postgraduate students through consultations, debugging support, milestone feedback, project-oriented learning, and assessment guidance.
- Experience translating mathematical, quantum-computing, machine-learning, and software concepts for audiences with varied technical backgrounds.

ACADEMIC AND PROFESSIONAL SERVICE

- Peer reviewer for *Future Generation Computer Systems*, *Neurocomputing*, and *IEEE Transactions on Vehicular Technology*.
- Committee Member and Quantum Machine Learning Tutor, ATIS 2025, Deakin University. [Committee](#) | [Program](#)
- Session Chair, QUICK 2025, Vellore Institute of Technology, India.
- Recipient of Elsevier reviewer recognition / reviewer excellence certificate, 2025. [Evidence](#)
- Volunteer Discipline Support Leader and student mentor, Charles Sturt University, 2019-2021.

INDUSTRY SOFTWARE EXPERIENCE

- Software Engineer and Developer Intern** - *Dynamic Sports Solutions / Trainer.co, Australia* **Oct-Dec 2021**
- Contributed to front-end and back-end development using PHP Laravel, JavaScript, HTML, Node.js, Linux, GitLab/GitHub, Trello, and Slack.
- Software Engineer and Developer Intern** - *LB Finance PLC, Sri Lanka* **Mar-Sep 2016**
- Developed, tested, and documented role-based employee access-management and helpdesk functionality supporting approximately 5,000 employees across 156 outlets.

AWARDS AND HONOURS

- Deakin University Postgraduate Research Scholarship 2022-2026.
- Postgraduate University Medal, Charles Sturt University, 2021.
- Executive Dean's Award, Charles Sturt University, 2020.
- Academic Achievement Award and Academic Merit Award, Charles Sturt University, 2019.
- Second Class Upper Division, BSc (Hons) in Information Technology, University of Moratuwa, 2017

REFEREES

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